seen spin-up and in the other they will have seen spin-down. But both worlds are indisputably *there*. If the question we're asking is "What is the chance I will end up being the experimenter on the spin-up branch of the wave function?," there doesn't seem to be any answer. You will not be one or other experimenters; your current single self will evolve, with certainty, into both of them. How are we supposed to talk about probabilities in such a situation?

It's a good question. To answer it, we have get a bit philosophical, and think about what "probability" really means.

o o o

You will not be surprised to learn that there are competing schools of thought on the issue of probability. Consider tossing a fair coin. "Fair" means that the coin will come up heads 50 percent of the time and tails 50 percent of the time. At least in the long run; nobody is surprised when you toss a coin twice and it comes up tails both times.

This "in the long run" caveat suggests a strategy for what we might mean by probability. For just a few coin tosses, we wouldn't be surprised at almost any outcome. But as we do more and more, we expect the total proportion of heads to come closer to 50 percent. So perhaps we can define the probability of getting heads as the fraction of times we actually would get heads, if the coin were tossed an infinite number of times.

This notion of what we mean by probability is sometimes called *frequentism*, as it defines probability as the relative frequency of an occurrence in a very large number of trials. It matches pretty well with our intuitive notions of how probability functions when we toss coins, roll dice, or play cards. To a frequentist, probability is an objective notion, since it only depends on features of the coin (or whatever other system we're talking about), not on us or our state of knowledge.

Frequentism fits comfortably with the textbook picture of quantum mechanics and the Born rule. Maybe you don't actually send an infinite number of electrons through a magnetic field to measure their spins, but you could send a very large number. (The Stern-Gerlach experiment is a favorite one to reproduce in undergraduate lab courses for physics majors, so over the years quite a number of spins have been measured this way.) We